From a language model to a planning agent

Weekend 1: Large language models

Carlos Cotrini
AI in Industry, ETH Zurich
Weekend 1, October 2026

From a language model to a planning agent

Language model
continues text

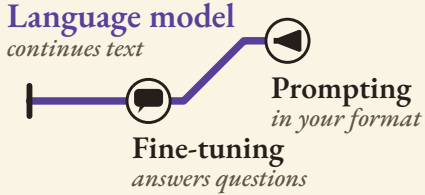
From a language model to a planning agent

Language model
continues text

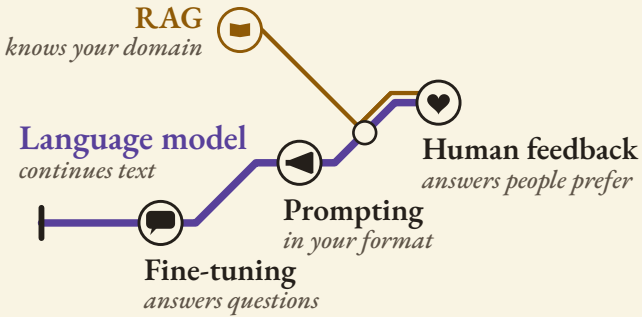


Fine-tuning
answers questions

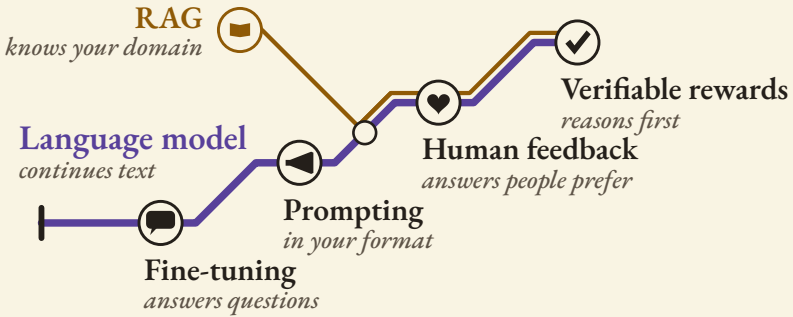
From a language model to a planning agent



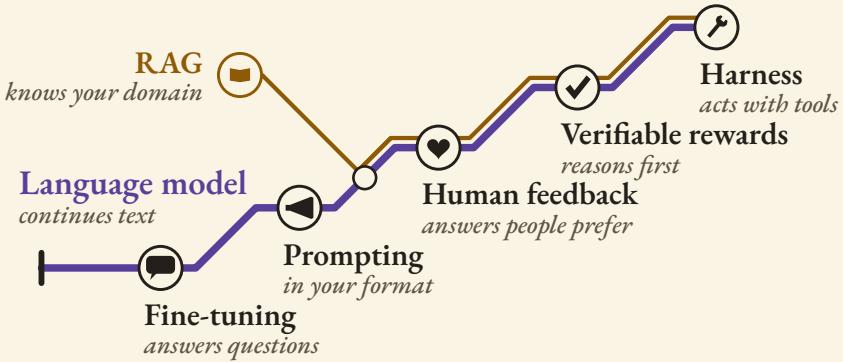
From a language model to a planning agent



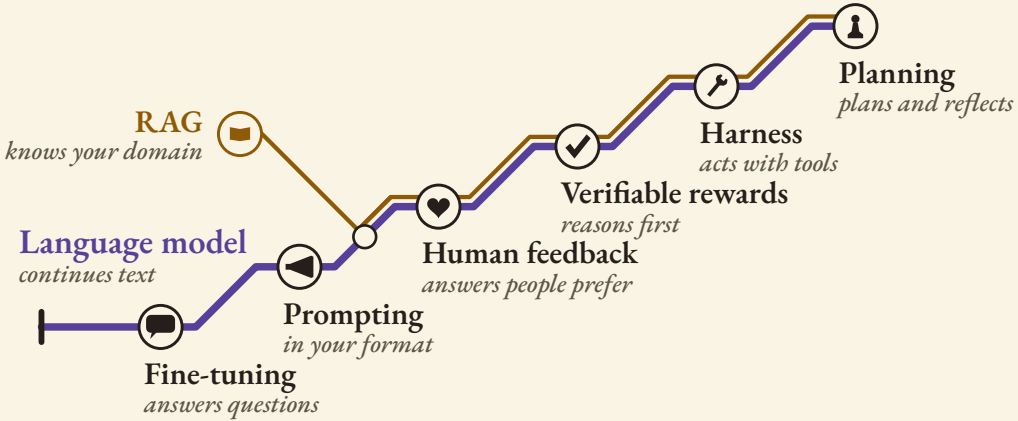
From a language model to a planning agent



From a language model to a planning agent

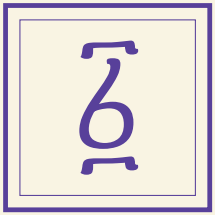


From a language model to a planning agent



From a language model to a planning agent





Section 1

Supervised fine-tuning

A pretrained model continues your text

You write

Dear customer, thank you for



It writes next

your order. We are pleased to confirm that it ships on Monday.

Invented examples

A pretrained model continues your text

You write	*It writes next*
Dear customer, thank you for	your order. We are pleased to confirm that it ships on Monday.
The central bank raised interest rates on	Thursday, citing inflation that stays above its target.

Invented examples

A pretrained model continues your text

You write	*It writes next*
Dear customer, thank you for	your order. We are pleased to confirm that it ships on Monday.
The central bank raised interest rates on	Thursday, citing inflation that stays above its target.
What is the capital of Switzerland?	What is the capital of Germany? What is the capital of Italy?

Invented examples

A pretrained model continues your text

You write	*It writes next*
Dear customer, thank you for	your order. We are pleased to confirm that it ships on Monday.
The central bank raised interest rates on	Thursday, citing inflation that stays above its target.
What is the capital of Switzerland?	What is the capital of Germany? What is the capital of Italy?

It continues the kind of page it thinks it is on.

Invented examples

People write the answers we want

Prompt *An invented example*

Explain what a mortgage is to a teenager.

Reply, written by a person

A mortgage is a loan for buying a home. You pay it back in small amounts every month.

People write the answers we want

Prompt *An invented example*

Explain what a mortgage is to a teenager.

Reply, written by a person

A mortgage is a loan for buying a home. You pay it back in small amounts every month.

And many more examples like it

People write the answers we want

Prompt *An invented example*

Explain what a mortgage is to a teenager.

Reply, written by a person

A mortgage is a loan for buying a home. You pay it back in small amounts every month.

And many more examples like it



People write the replies

People write the answers we want

Prompt *An invented example*

Explain what a mortgage is to a teenager.

Reply, written by a person

A mortgage is a loan for buying a home. You pay it back in small amounts every month.

And many more examples like it



People write the replies

InstructGPT, 2022: 12 725 prompts, replies by about 40 contractors. LIMA, 2023: 1 000 examples.

Game 3's learning, on new text

What the model has so far

Explain what a mortgage is to a teenager. **A mortgage is a**

The next word of the reply

loan

Game 3's learning, on new text

What the model has so far

Explain what a mortgage is to a teenager. **A mortgage is a**

The next word of the reply

loan

Before



Invented chances, for illustration

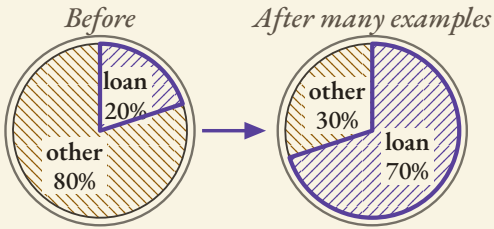
Game 3's learning, on new text

What the model has so far

Explain what a mortgage is to a teenager. **A mortgage is a**

The next word of the reply

loan



Invented chances, for illustration

Game 3's learning, on new text

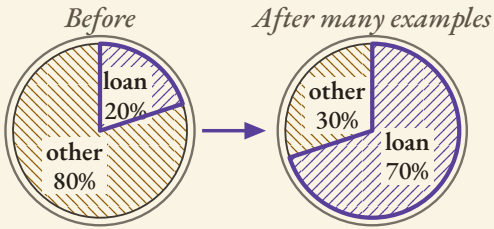
What the model has so far

Explain what a mortgage is to a teenager. **A mortgage is a**

The next word of the reply

loan

Game 3's training: what came before, and the word that came next.



Invented chances, for illustration

A conversation is one long prefix

What you see

System You are a helpful assistant of our bank.

You Explain what a mortgage is to a teenager.

Assistant

A conversation is one long prefix

What you see

System You are a helpful assistant of our bank.

You Explain what a mortgage is to a teenager.

Assistant

What the model reads

`[system]` You are a helpful assistant of our bank.
`[user]` Explain what a mortgage is to a teenager.
`[assistant]`



The markers differ from one model to another.

A conversation is one long prefix

What you see

System	You are a helpful assistant of our bank.
You	Explain what a mortgage is to a teenager.
Assistant	A mortgage is a loan for buying a home.

What the model reads

`[system]` You are a helpful assistant of our bank.
`[user]` Explain what a mortgage is to a teenager.
`[assistant]` **A mortgage is a loan for buying a home.** `[end]`



The markers differ from one model to another.

A conversation is one long prefix

What you see

System	You are a helpful assistant of our bank.
You	Explain what a mortgage is to a teenager.
Assistant	A mortgage is a loan for buying a home.

What the model reads

`[system]` You are a helpful assistant of our bank.
`[user]` Explain what a mortgage is to a teenager.
`[assistant]` **A mortgage is a loan for buying a home.** `[end]`



The markers differ from one model to another.



Every new turn sends the whole conversation again.

Fine-tuning teaches a manner of answering

You ask

What is our refund policy?

The chat model answers

Thank you for asking! Our policy is simple: you can return any item within a month for a full refund.

An invented answer

Fine-tuning teaches a manner of answering

You ask

What is our refund policy?

The chat model answers

Thank you for asking! Our policy is simple: you can return any item within a month for a full refund.

An invented answer

The manner. *Polite, short, in the format you expect.*

Fine-tuning teaches a manner of answering

You ask

What is our refund policy?

The chat model answers

Thank you for asking! Our policy is simple: you can return any item within a month for a full refund.

An invented answer

The manner. *Polite, short, in the format you expect.*

The facts. *The model never saw our policy. It writes the words that sound right.*

Fine-tuning teaches a manner of answering

You ask

What is our refund policy?

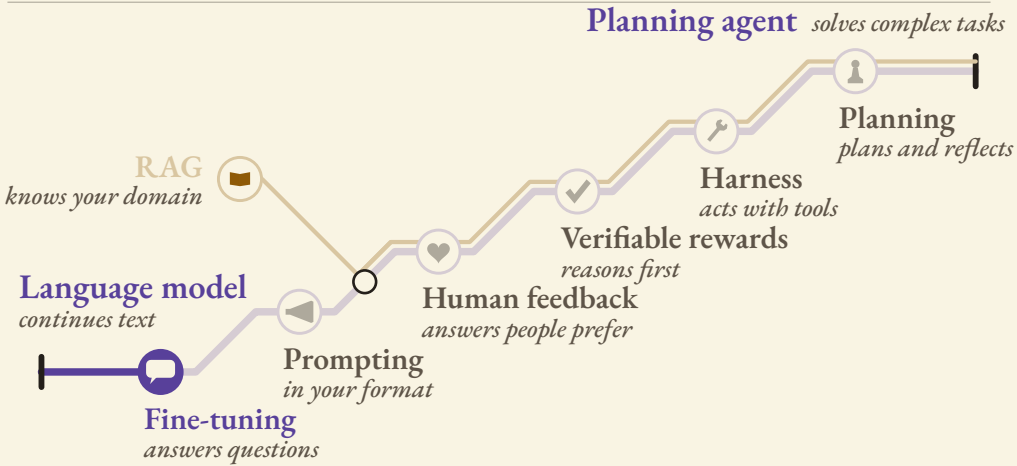
The chat model answers

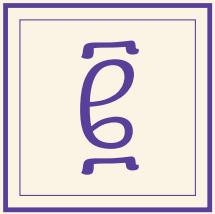
Thank you for asking! Our policy is simple: you can return any item within a month for a full refund.

An invented answer

- ***The manner.*** *Polite, short, in the format you expect.*
- ***The facts.*** *The model never saw our policy. It writes the words that sound right.*
- *Facts come from pretraining, or from documents you give it.*

One rung up: a chat model





Section 2

Prompt engineering

Different words in, a different answer out

You ask

Explain what a mortgage is.



The same model answers



Invented answers

Different words in, a different answer out

You ask

The same model answers

Explain what a mortgage is. →



Explain what a mortgage is to a teenager, in two sentences. →

A mortgage is a loan for a home. You pay it back monthly.

Invented answers

Different words in, a different answer out

You ask

Explain what a mortgage is.

Explain what a mortgage is to a teenager, in two sentences.

Explain what a mortgage is, as a table with two columns.

The same model answers



A mortgage is a loan for a home. You pay it back monthly.

Term	Meaning
Loan	Money you borrow
Interest	The price of the loan

Invented answers

Different words in, a different answer out

You ask

Explain what a mortgage is.

Explain what a mortgage is to a teenager, in two sentences.

Explain what a mortgage is, as a table with two columns.

The same model answers



A mortgage is a loan for a home. You pay it back monthly.

Term	Meaning
Loan	Money you borrow
Interest	The price of the loan

Invented answers

The model is the same in all three rows. Only the words that go in are different.

A prompt is a brief to a new colleague

Task Explain what a mortgage is.

A capable new colleague has read everything and knows nothing about your company.

A prompt is a brief to a new colleague

Role	You are a bank advisor.
Task	Explain what a mortgage is.



A capable new colleague has read everything and knows nothing about your company.

A prompt is a brief to a new colleague

Role	You are a bank advisor.
Task	Explain what a mortgage is.
Example	Like this: a loan is money you borrow and pay back.

A capable new colleague has read everything and knows nothing about your company.

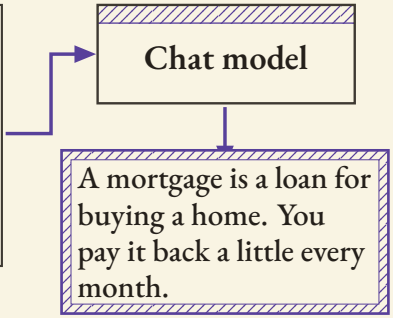
A prompt is a brief to a new colleague

Role	You are a bank advisor.
Task	Explain what a mortgage is.
Example	Like this: a loan is money you borrow and pay back.
Format	Two sentences, in plain words.

A capable new colleague has read everything and knows nothing about your company.

A prompt is a brief to a new colleague

Role	You are a bank advisor.
Task	Explain what a mortgage is.
Example	Like this: a loan is money you borrow and pay back.
Format	Two sentences, in plain words.



A capable new colleague has read everything and knows nothing about your company.

An invented answer

Examples in the prompt teach by showing

The prompt

Message The parcel arrived broken. *Label* Complaint

Examples in the prompt teach by showing

The prompt

Message The parcel arrived broken. *Label* Complaint

Message Thank you, great service! *Label* Praise

Examples in the prompt teach by showing

The prompt

Message The parcel arrived broken. *Label* Complaint

Message Thank you, great service! *Label* Praise

Message Where is my order? *Label*

Examples in the prompt teach by showing

The prompt

Message The parcel arrived broken. *Label* Complaint

Message Thank you, great service! *Label* Praise

Message Where is my order? *Label*

Question

The model draws

Examples in the prompt teach by showing

The prompt

Message The parcel arrived broken. *Label* Complaint

Message Thank you, great service! *Label* Praise

Message Where is my order? *Label*

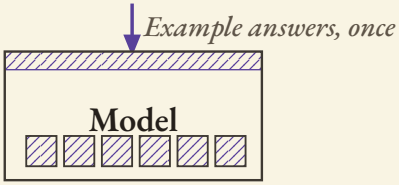
Question
The model draws



Nothing inside the model has changed. The examples have to be in the prompt every time.

A prompt steers the model without changing it

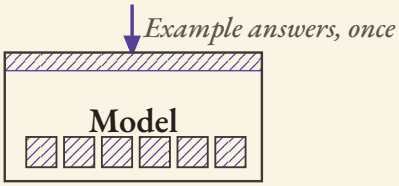
Fine-tuning



The numbers inside change, and so do all later answers.

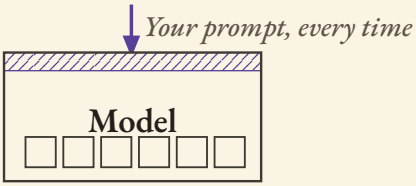
A prompt steers the model without changing it

Fine-tuning



The numbers inside change, and so do all later answers.

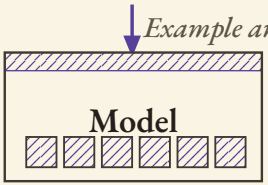
Prompting



The numbers inside stay the same. Only the input differs.

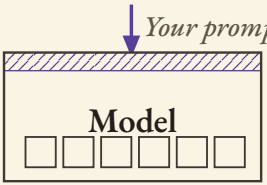
A prompt steers the model without changing it

Fine-tuning



The numbers inside change, and so do all later answers.

Prompting

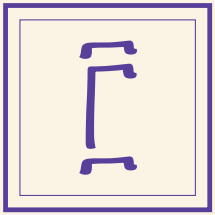


The numbers inside stay the same. Only the input differs.

Prompting is quick to try and training is slow. Start with the prompt, and train when it is not enough.

Two rungs up: your words steer the model

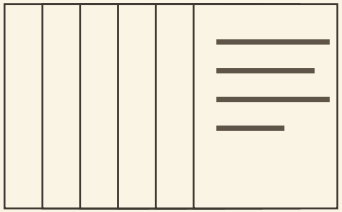




Section 3

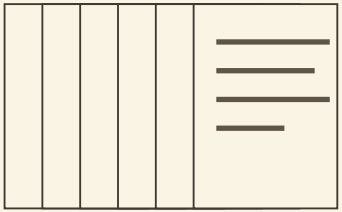
RAG

A prompt holds only part of the archive



All our documents

A prompt holds only part of the archive

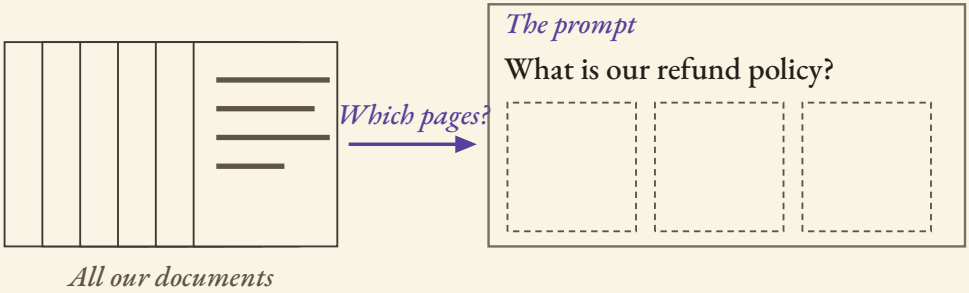


All our documents

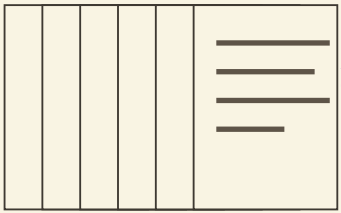
The prompt

What is our refund policy?

A prompt holds only part of the archive



A prompt holds only part of the archive



All our documents

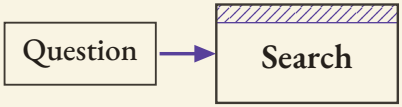
Which pages?

The prompt

What is our refund policy?

In one study, the right pages added facts better than fine-tuning.

Search first, then answer from what was found



Before anyone asks, every page becomes a row, as in Game 4.

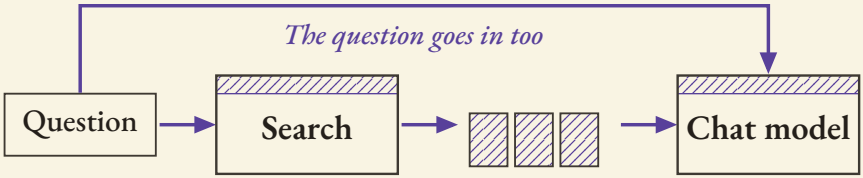
Search first, then answer from what was found



Before anyone asks, every page becomes a row, as in Game 4.

The question becomes a row too. The closest rows are the best pages.

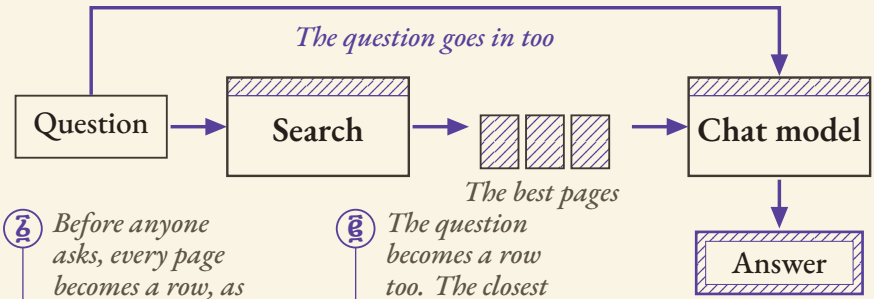
Search first, then answer from what was found



Before anyone asks, every page becomes a row, as in Game 4.

The question becomes a row too. The closest rows are the best pages.

Search first, then answer from what was found



The answer can point to its source

What is our refund policy?

You can return any item within a month for a full refund.

An invented answer

The answer can point to its source

What is our refund policy?

You can return any item within a month for a full refund.



Source: refund policy, returns

The facts come from your pages, so you can check them.

An invented answer

The answer can point to its source

What is our refund policy?

You can return any item within a month for a full refund.



Source: refund policy, returns

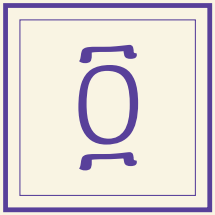
The facts come from your pages, so you can check them.

The search can miss the right page. Check the source.

An invented answer

RAG runs beside the line

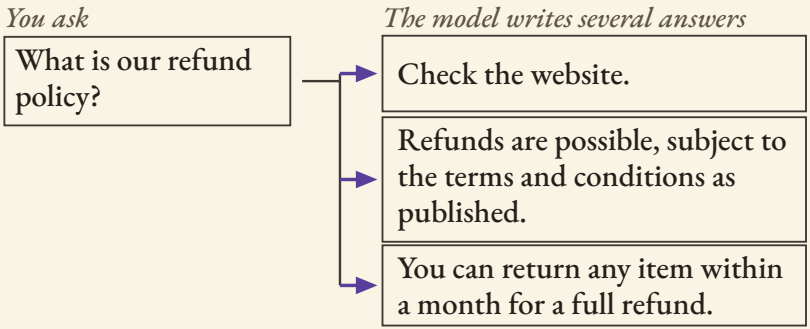




Section 4

Human feedback

People rank the answers



Invented answers

People rank the answers

You ask

What is our refund policy?



A person ranks them

The model writes several answers

Check the website.

Refunds are possible, subject to the terms and conditions as published.

You can return any item within a month for a full refund.

Second

Third

Best

Invented answers

People rank the answers

You ask

What is our refund policy?

The model writes several answers

- Check the website. — Second
- Refunds are possible, subject to the terms and conditions as published. — Third
- You can return any item within a month for a full refund. — Best

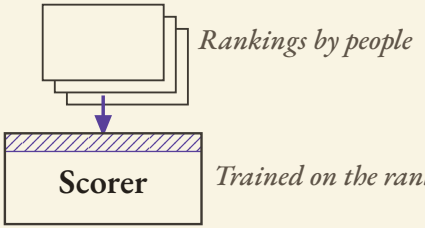


A person ranks them

Invented answers

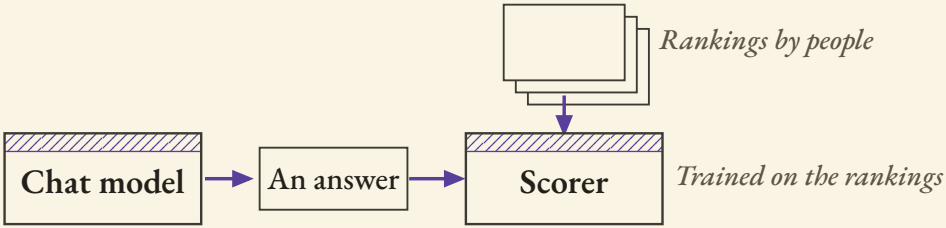
InstructGPT, 2022: 33 207 questions, each with 4 to 9 answers ranked.

A scorer learns the people's taste



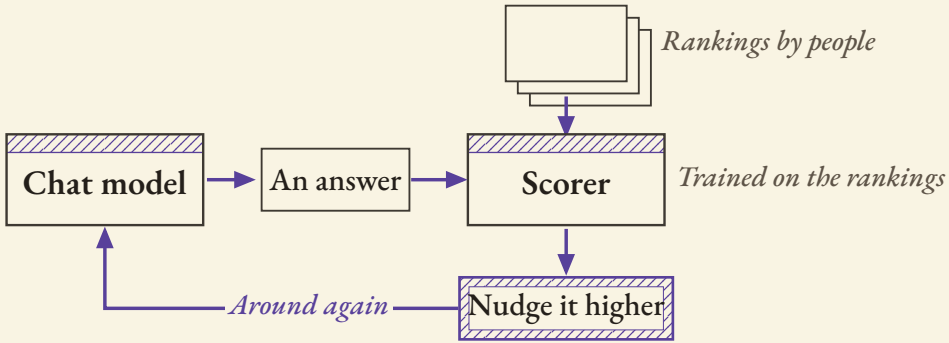
Invented example

A scorer learns the people's taste



Invented example

A scorer learns the people's taste



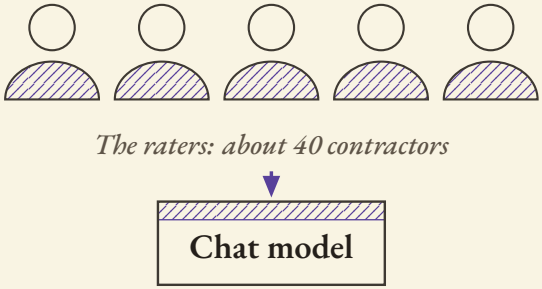
Invented example

The model learns the taste of its raters

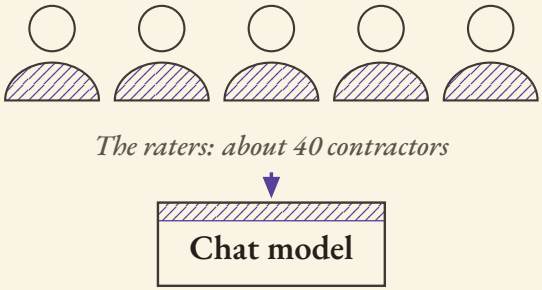


The raters: about 40 contractors

The model learns the taste of its raters



The model learns the taste of its raters

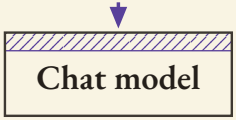


In the study, the raters preferred the answers of a model 100 times smaller.

The model learns the taste of its raters



The raters: about 40 contractors

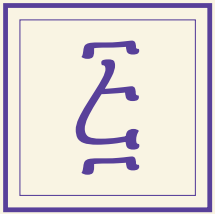


In the study, the raters preferred the answers of a model 100 times smaller.

Approval can reward flattery: people sometimes prefer a convincing, agreeable answer to a correct one.

Three rungs up: answers people prefer



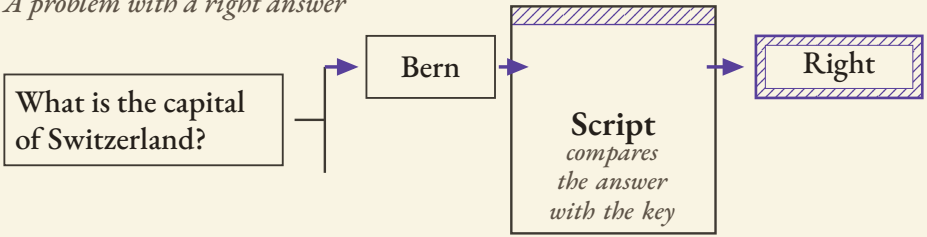


Section 5

Verifiable rewards

A script can check the answer

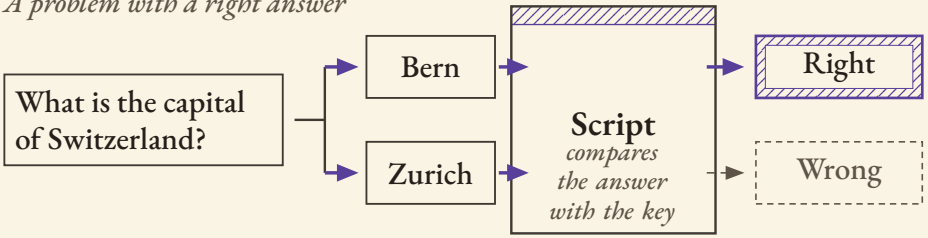
A problem with a right answer



Invented answers

A script can check the answer

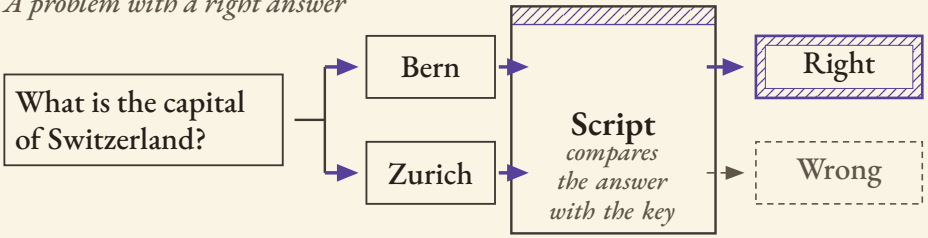
A problem with a right answer



Invented answers

A script can check the answer

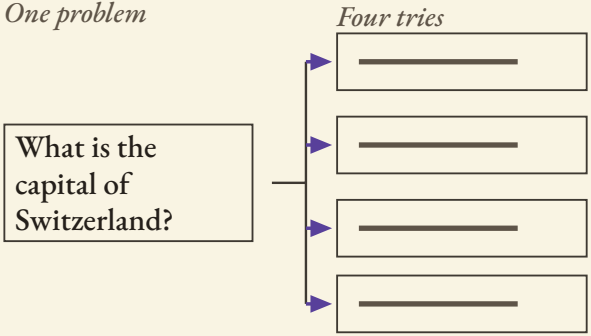
A problem with a right answer



For code, the script runs the tests. A program that passes them is right.

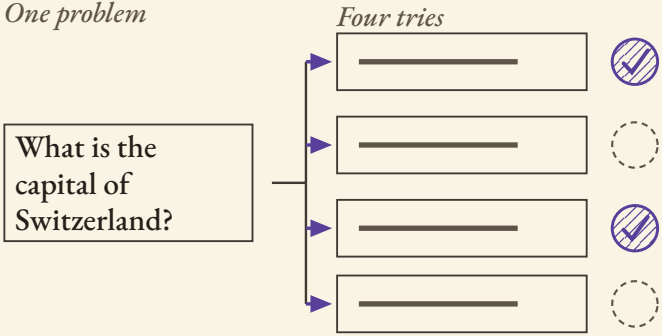
Invented answers

The right answers become more likely



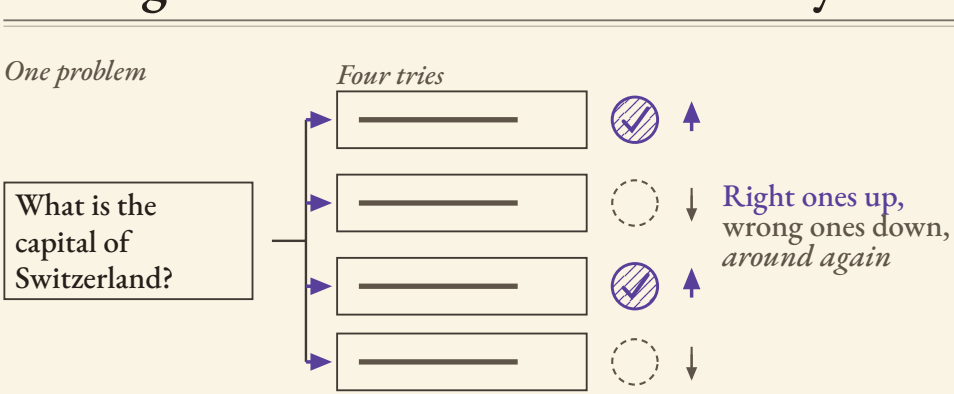
Invented example

The right answers become more likely



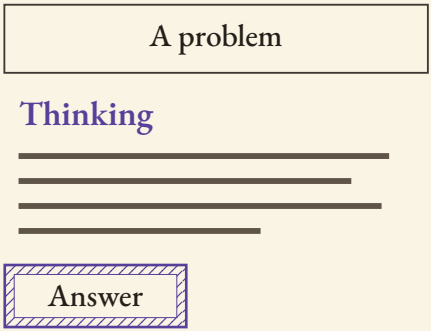
Invented example

The right answers become more likely



Invented example

The model learns to think before it answers



The model learns to think before it answers

A problem

Thinking

Answer

Right answers on a maths competition

Before training 15.6%

After training 77.9%

The model learns to think before it answers

A problem

Thinking

Answer

Nobody wrote the steps. They appeared because they led to more right answers.

Right answers on a maths competition

Before training 15.6%

After training 77.9%

Four rungs up: it reasons before it answers



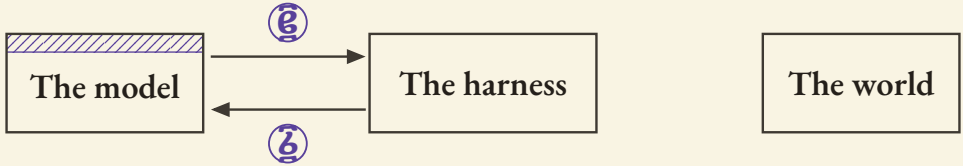
6

Section 6

A harness

The harness turns text into actions

The program around the model. It turns text into actions and feeds back what happened.

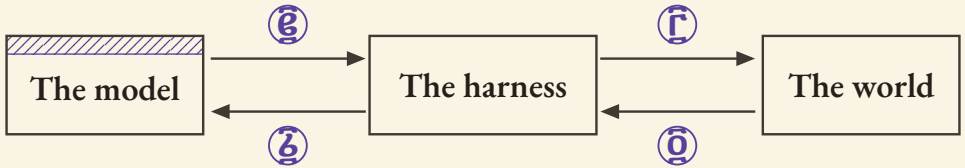


- Builds the prompt: the tools and the transcript
- Reads the reply and finds the tool call

Anthropic, tool use, 2026

The harness turns text into actions

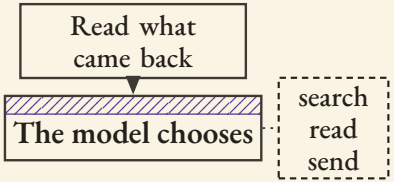
The program around the model. It turns text into actions and feeds back what happened.



- Builds the prompt: the tools and the transcript
- Reads the reply and finds the tool call
- Runs that tool on the world
- Appends the result, then calls the model again or stops

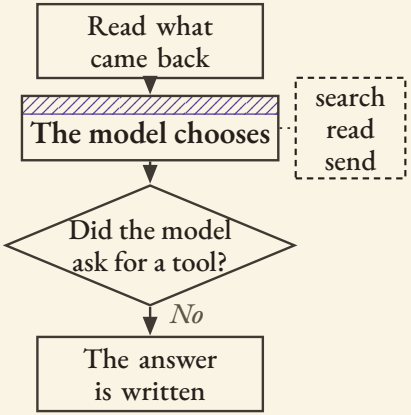
Anthropic, tool use, 2026

The loop: think, act, observe, again



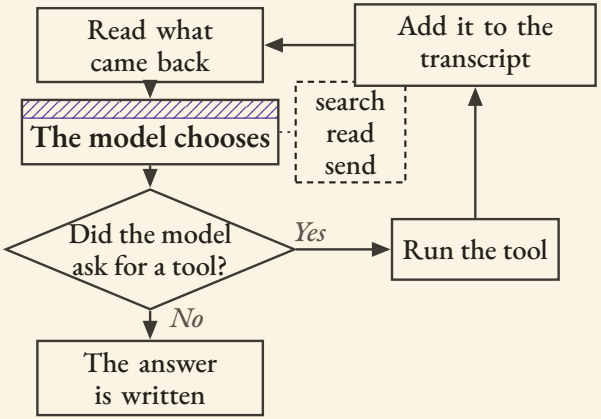
Yao and others, ReAct, ICLR 2023

The loop: think, act, observe, again



Yao and others, ReAct, ICLR 2023

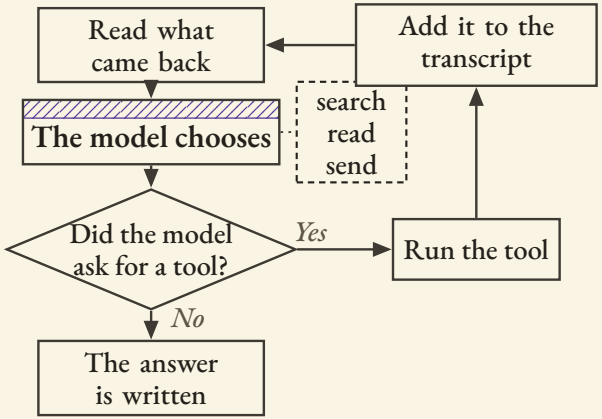
The loop: think, act, observe, again

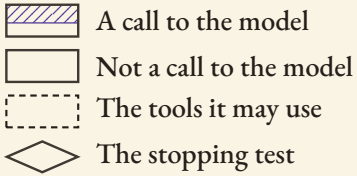


The stopping test is in the loop's own condition. It is not a step counter.

Yao and others, ReAct, ICLR 2023

The loop: think, act, observe, again





The stopping test is in the loop's own condition. It is not a step counter.

Yao and others, ReAct, ICLR 2023

The transcript the model is called with

The transcript

Request

Our refund policy, please.

This box and the prompt are the context window

Written once, by a person

You are the support agent of our shop. Tools: search, read, send. One tool call per turn.

Invented example

The transcript the model is called with

The transcript

Request

Our refund policy, please.

Action

search(“refund policy”)

This box and the prompt are the context window

Written once, by a person

You are the support agent of our shop. Tools: search, read, send. One tool call per turn.

What the model wrote this step

search(“refund policy”)

Invented example

The transcript the model is called with

The transcript

Request	Our refund policy, please.
Action	search(“refund policy”)
Result	Returns: any item within one month, full refund.
Action	reply(“You can return any item within a month.”)

This box and the prompt are the context window

Written once, by a person

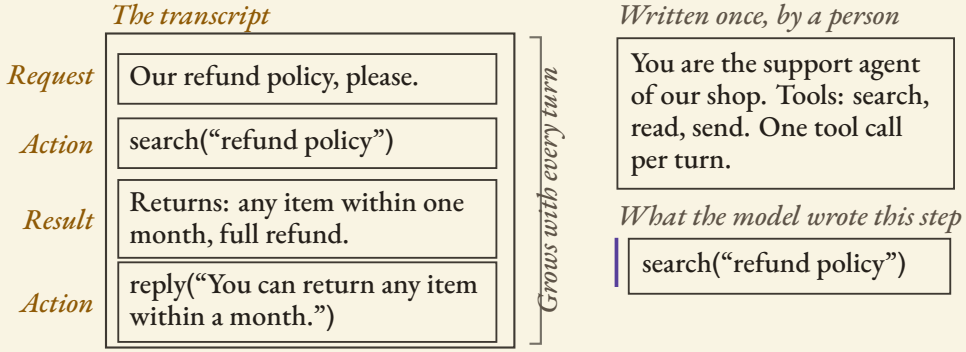
You are the support agent of our shop. Tools: search, read, send. One tool call per turn.

What the model wrote this step

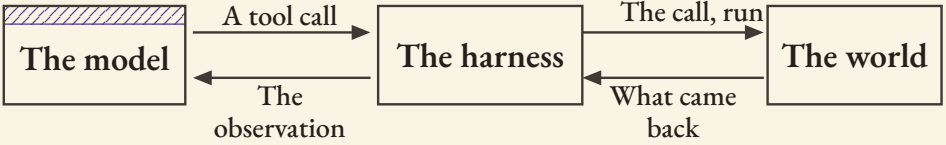
search(“refund policy”)

Invented example

The transcript the model is called with

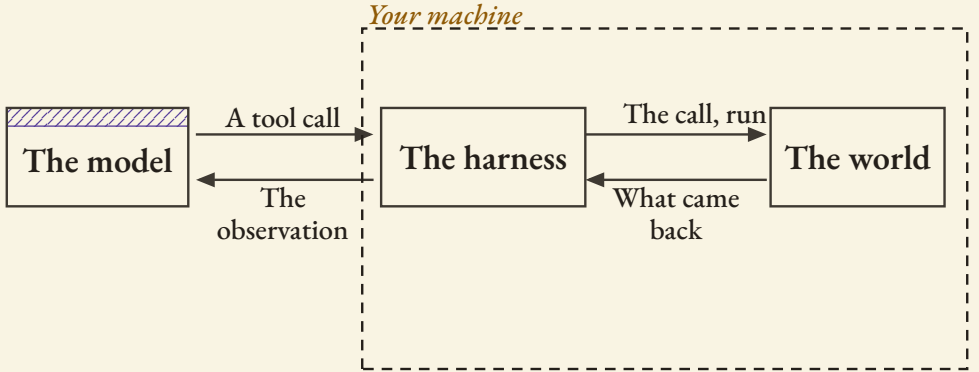


An agent is a system



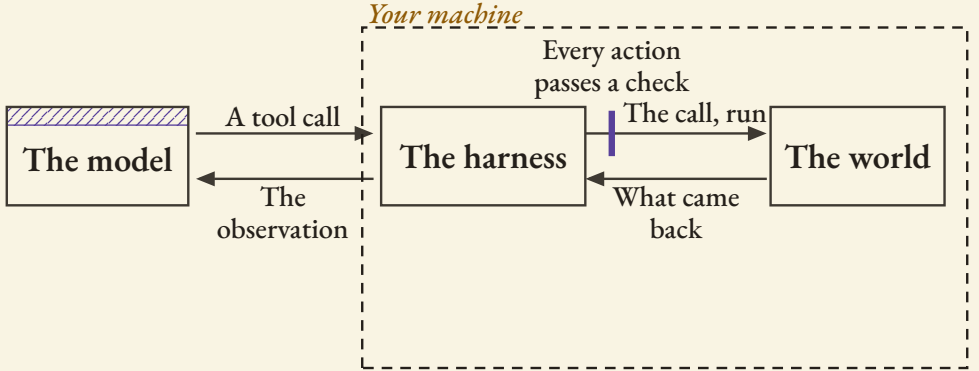
Cherny, Claude Code, Anthropic, 2025

An agent is a system



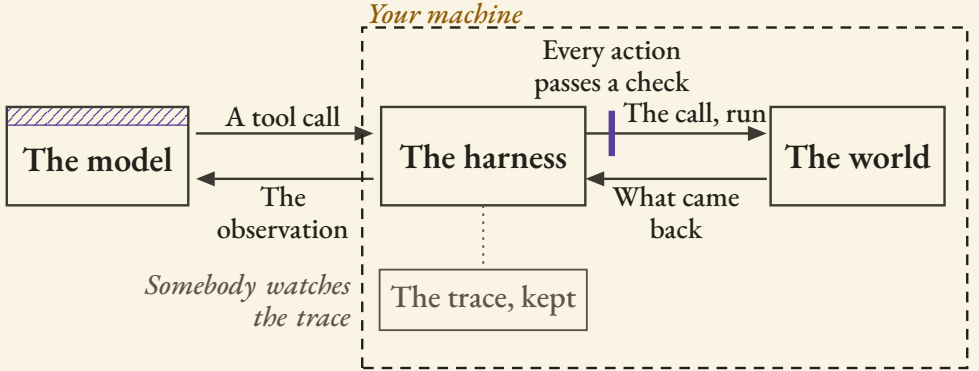
Cherny, Claude Code, Anthropic, 2025

An agent is a system



Cherny, Claude Code, Anthropic, 2025

An agent is a system



Cherny, Claude Code, Anthropic, 2025

Five rungs up: it acts with tools



7

Section 7

Planning strategies

These tasks are games

The task	A position	A move	You have won when
Chess	The board right now	A legal move	Checkmate
Fixing a bug	The repository, and a failing test	An edit, or a command	The tests pass
Booking a table	The page you are on	A click, or something typed	The booking is confirmed
Buying the right item	The search results	A search, or a click on a product	It matches what was asked

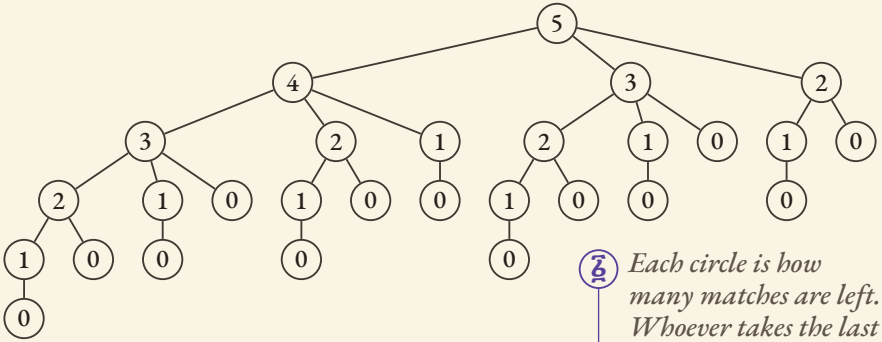
These tasks are games

The task	A position	A move	You have won when
Chess	The board right now	A legal move	Checkmate
Fixing a bug	The repository, and a failing test	An edit, or a command	The tests pass
Booking a table	The page you are on	A click, or something typed	The booking is confirmed
Buying the right item	The search results	A search, or a click on a product	It matches what was asked

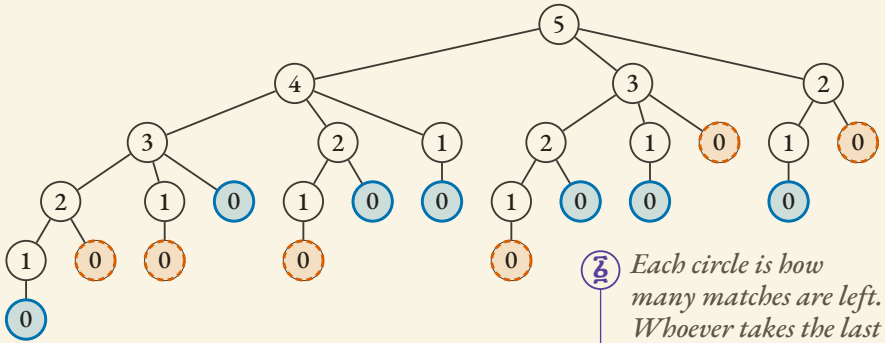


In all four, one move is rarely right or wrong on its own. It is right if where it leads is good.

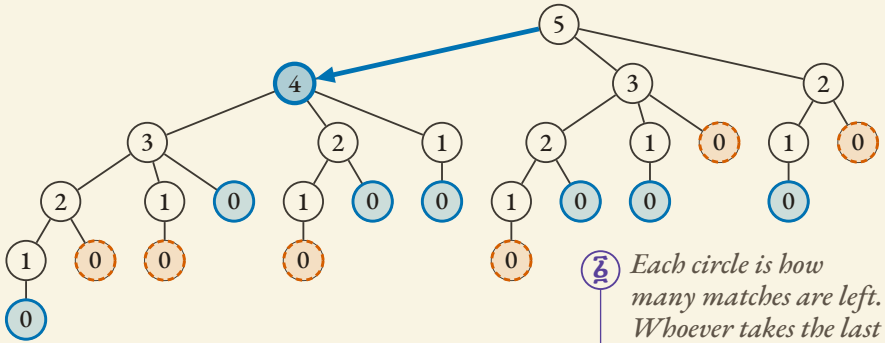
The whole game tree, for once



The whole game tree, for once

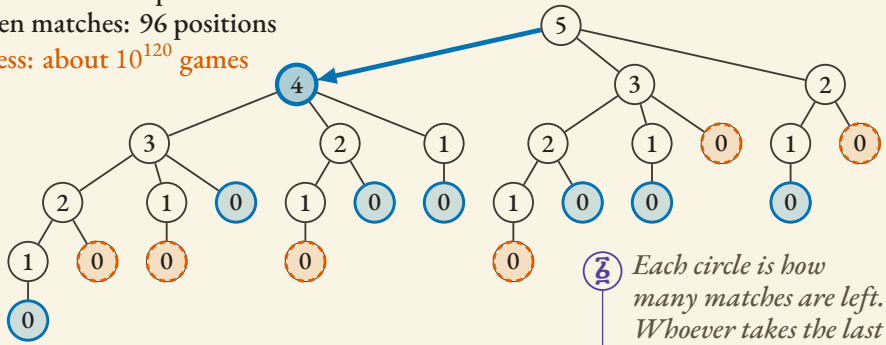


The whole game tree, for once



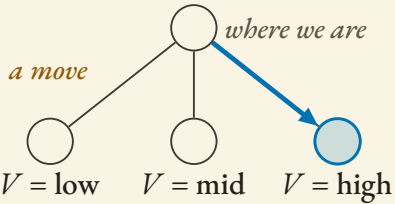
The whole game tree, for once

Five matches: 28 positions
Seven matches: 96 positions
Chess: about 10^{120} games



Two ways to score a position

A critic: an opinion

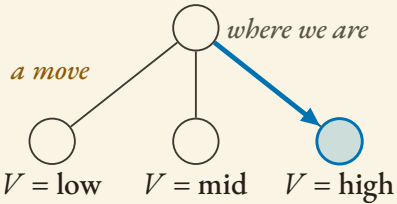


A critic is fast, and it can be confidently wrong.

X to move

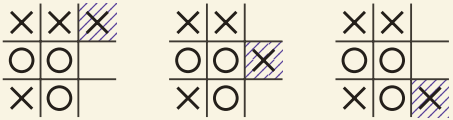
Two ways to score a position

A critic: an opinion



A critic is fast, and it can be confidently wrong.

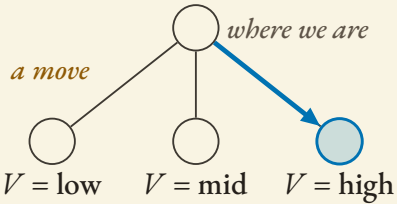
A rollout: play it out, at random



X to move

Two ways to score a position

A critic: an opinion



A critic is fast, and it can be confidently wrong.

A rollout: play it out, at random

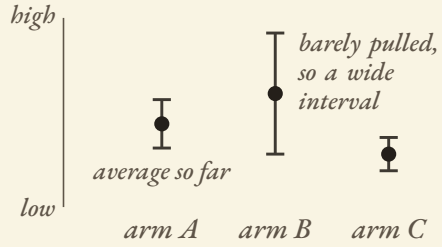


The average of 100 random playouts

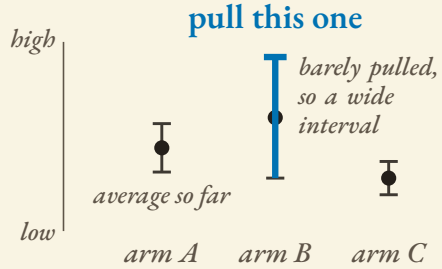
A real measurement, one noisy sample at a time.

X to move

Pick the arm with the highest upper bound

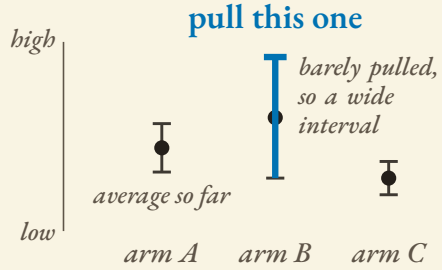


Pick the arm with the highest upper bound

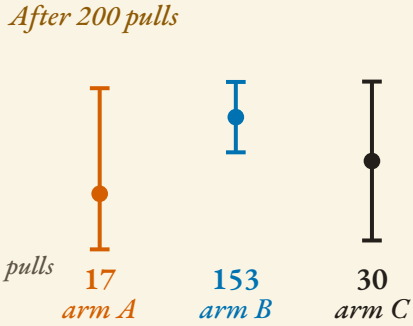


A barely pulled arm has a wide interval, so it is tried even when its average is poor.

Pick the arm with the highest upper bound

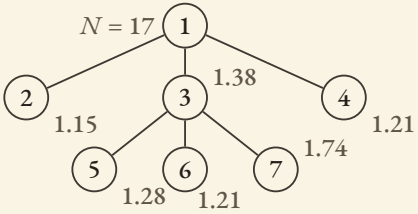


A barely pulled arm has a wide interval, so it is tried even when its average is poor.

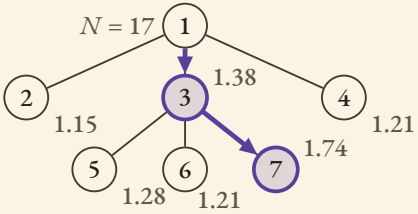


In the tree, this rule is UCT: how good it looked, plus how little we know.

The same loop, with a model in every box

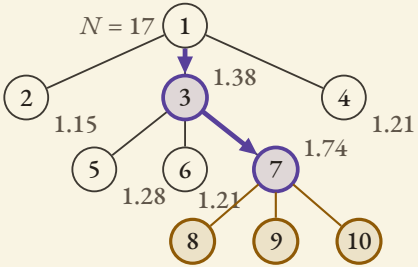


The same loop, with a model in every box



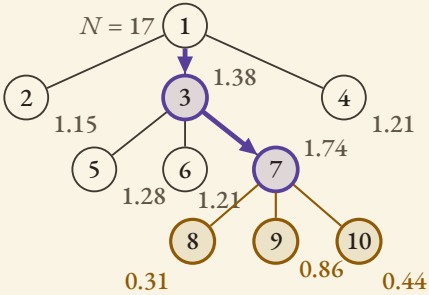
g̲ Selection *by UCT*

The same loop, with a model in every box



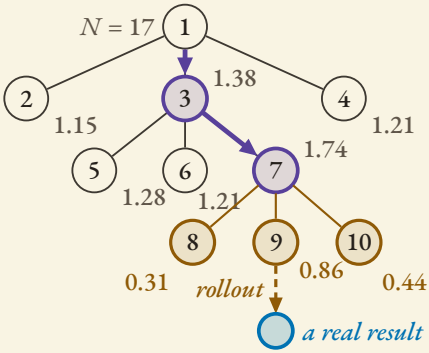
Selection	*by UCT*
Expansion	*the actor proposes*

The same loop, with a model in every box



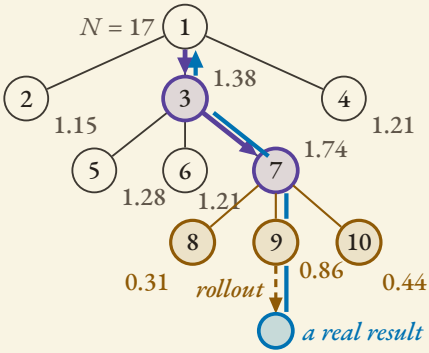
Selection	*by UCT*
Expansion	*the actor proposes*
Evaluation	*the critic scores*

The same loop, with a model in every box



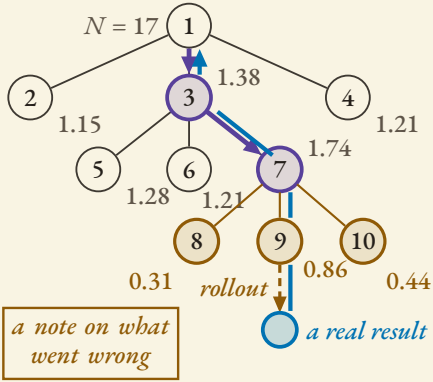
Selection	*by UCT*
Expansion	*the actor proposes*
Evaluation	*the critic scores*
Simulation	*the best one, to the end*

The same loop, with a model in every box



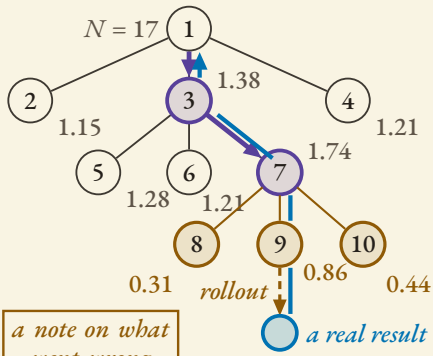
Selection	*by UCT*
Expansion	*the actor proposes*
Evaluation	*the critic scores*
Simulation	*the best one, to the end*
Backpropagation	*the real result, up*

The same loop, with a model in every box



1	**Selection**	*by UCT*
2	**Expansion**	*the actor proposes*
3	**Evaluation**	*the critic scores*
4	**Simulation**	*the best one, to the end*
5	**Backpropagation**	*the real result, up*
6	**Reflection**	*the model writes it down*

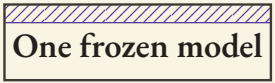
The same loop, with a model in every box



Selection	*by UCT*	
Expansion	*the actor proposes*	
Evaluation	*the critic scores*	
Simulation	*the best one, to the end*	
Backpropagation	*the real result, up*	
6 Reflection	*the model writes it down*	*into the next prompt*

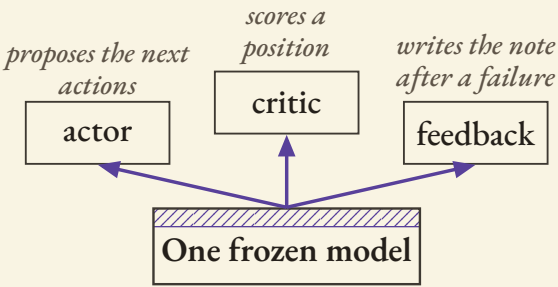
Grey: UCT, what selection compares.
Ochre: V from the critic, and all that is new.

Three jobs, one set of weights



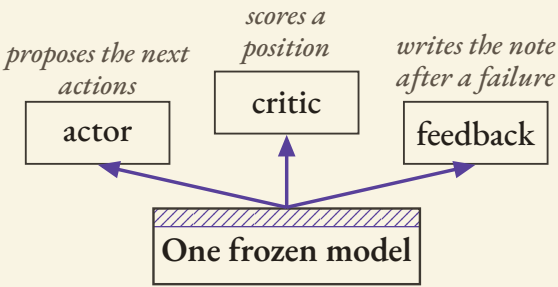
Zhou and others, LATS, ICML 2024

Three jobs, one set of weights



Zhou and others, LATS, ICML 2024

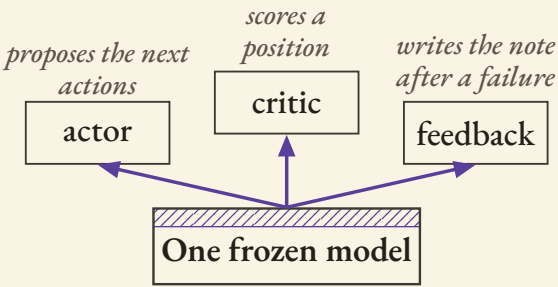
Three jobs, one set of weights



The same weights every time.
Only the prompt changes.

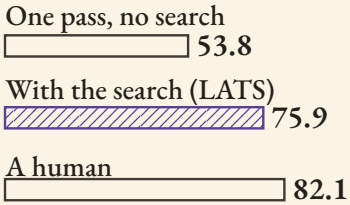
Zhou and others, LATS, ICML 2024

Three jobs, one set of weights



The same weights every time.
Only the prompt changes.

A shopping site, 1.18 million real products



Zhou and others, LATS, ICML 2024

Six rungs up: it plans and reflects

